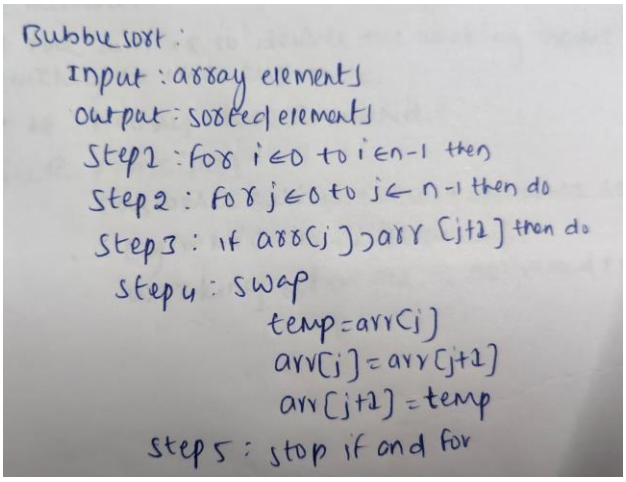
Practical 1

Aim:
To Implementation and Time analysis of sorting algorithms. Bubble sort, Selection sort, Insertion sort, Merge sort and Quicksort

Bubble sort

Algorithm:

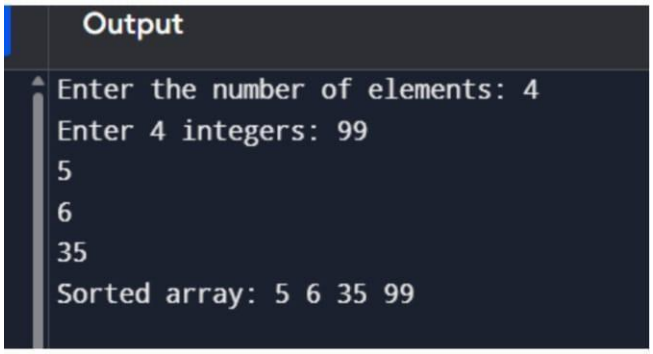


Code for bubble sort:

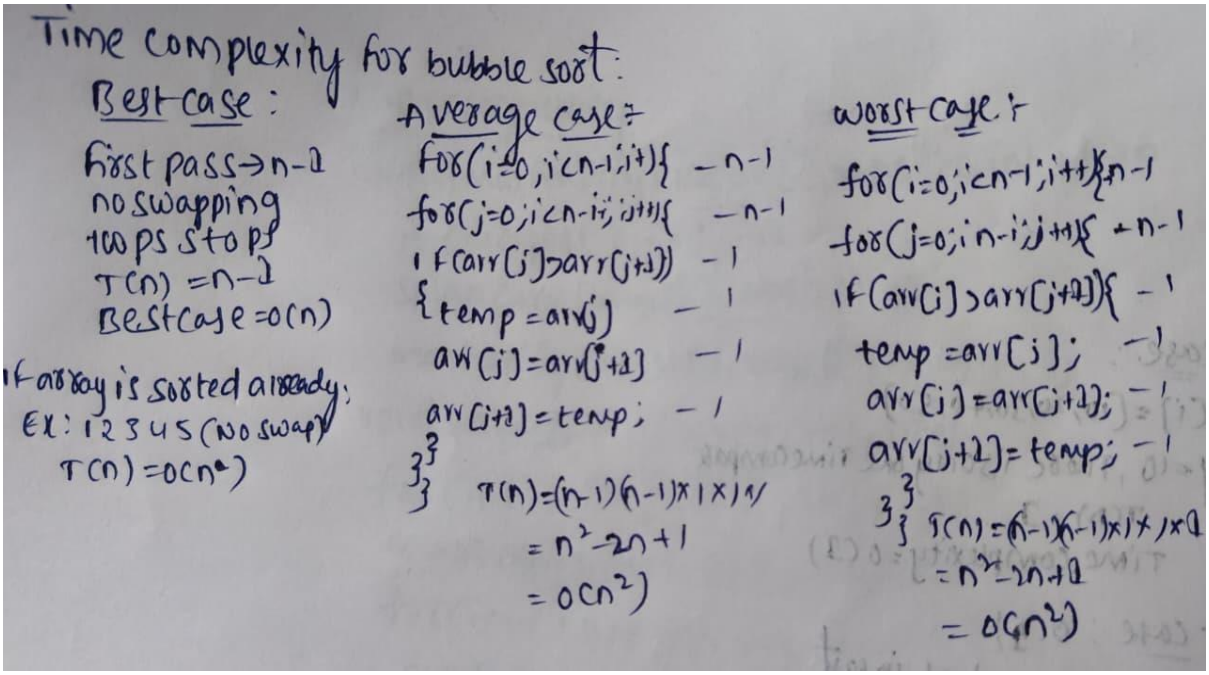
```
#include <iostream>
#include <vector>
using namespace std;
void bubbleSort(vector<int>& arr) {
    int n = arr.size();
    for (int i = 0; i < n - 1; ++i) {
        for (int j = 0; j < n - i - 1; ++j) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
            }
    }
```

```
int main() {
    int size;
    cout << "Enter the number of elements: ";
    cin >> size;
    vector<int> arr(size);
    cout << "Enter " << size << " integers: ";
    for (int i = 0; i < size; ++i) {
        cin >> arr[i];
    }
    bubbleSort(arr);
    cout << "Sorted array: ";
    for (int i = 0; i < size; ++i) {
        cout << arr[i] << " ";
    }
    cout << endl;
    return 0;
}
```

Output:

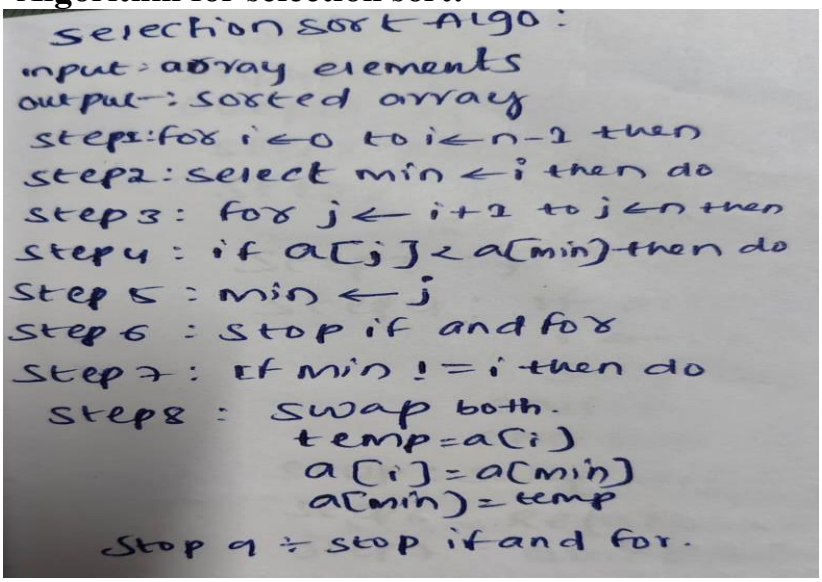


Time complexity:



Selection sort

Algorithm for selection sort:



Code for selection sort:

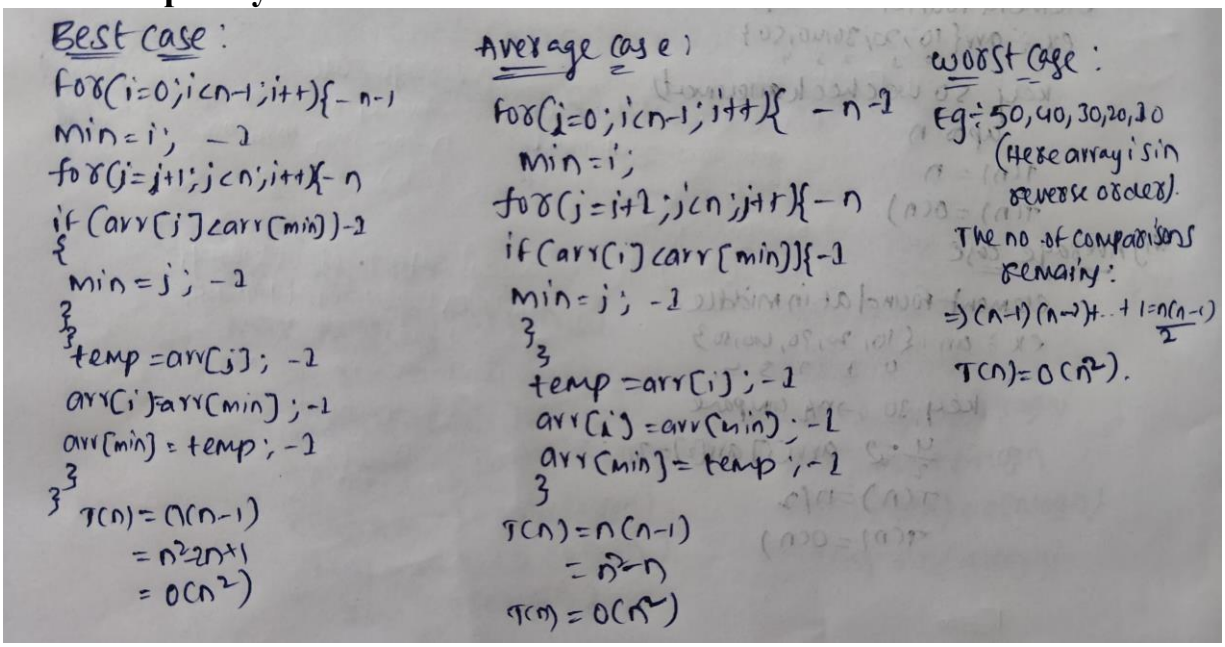
```
#include <iostream>
#include <vector>
void selectionSort(std::vector<int>& arr) {
    int n = arr.size();
    for (int i = 0; i < n - 1; ++i) {
        int minIndex = i;
        for (int j = i + 1; j < n; ++j) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }
        if (minIndex != i) {
            std::swap(arr[i], arr[minIndex]);
        }}
int main() {
     int n;
    std::cout << "Enter the number of elements: ";
    std::cin >> n;
    if (n <= 0) {
        return 0;
    }std::vector<int> arr(n);
```

```
    std::cout << "Enter " << n << " integers: ";
    for (int i = 0; i < n; ++i) {
        std::cin >> arr[i];
    }

    selectionSort(arr);

    std::cout << "Sorted array: ";
    for (int i = 0; i < n; ++i) {
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
    return 0;
}
```

Output:

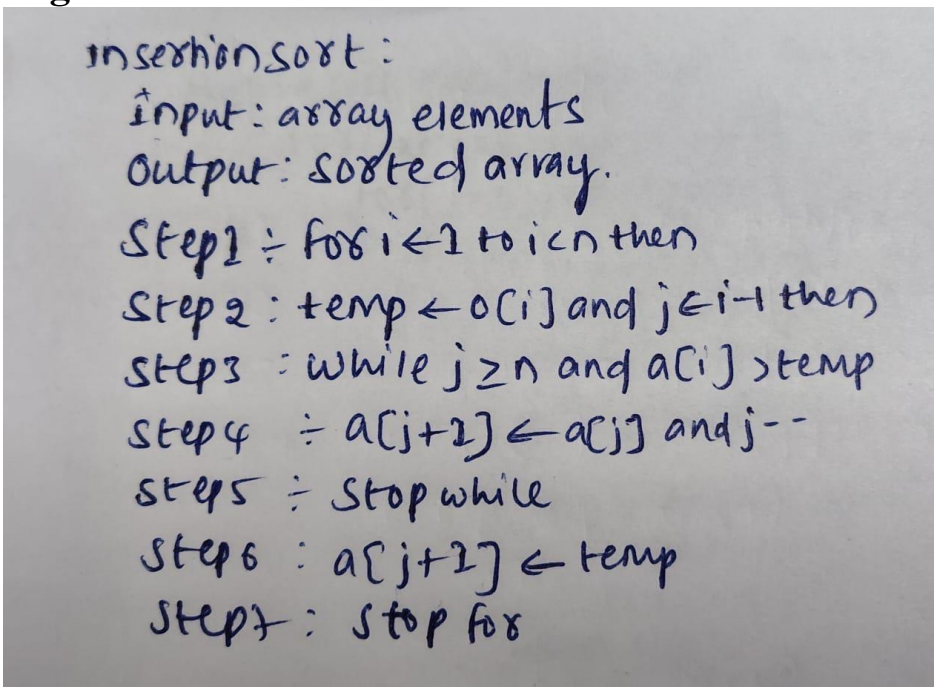
```
Enter the number of elements: 5
Enter 5 integers: 1
02
88
6
-8
Sorted array: -8 1 2 6 88
```

Time complexity:



Insertion sort

Algorithm:



Code:

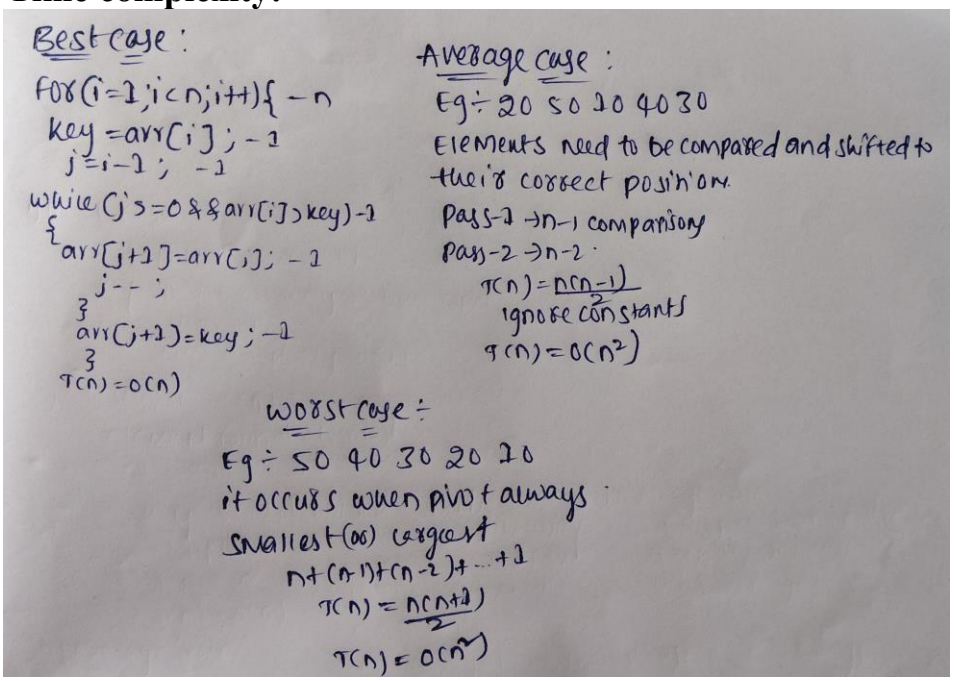
```
#include <iostream>
#include <vector>
void insertionSort(std::vector<int>& arr) {
    int n = arr.size();
    for (int i = 1; i < n; ++i) {
        int key = arr[i];
        int j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        arr[j + 1] = key;
    }
}
int main() {
    int n;
    std::cout << "Enter the number of elements: ";
    std::cin >> n;
    if (n <= 0)
        return0;}
```

```
    for (int i = 0; i < n; ++i) {
        std::cin >> arr[i];
    }

    insertionSort(arr);

    std::cout << "Sorted array: ";
    for (int i = 0; i < n; ++i) {
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
    return 0;
}
```

Output:

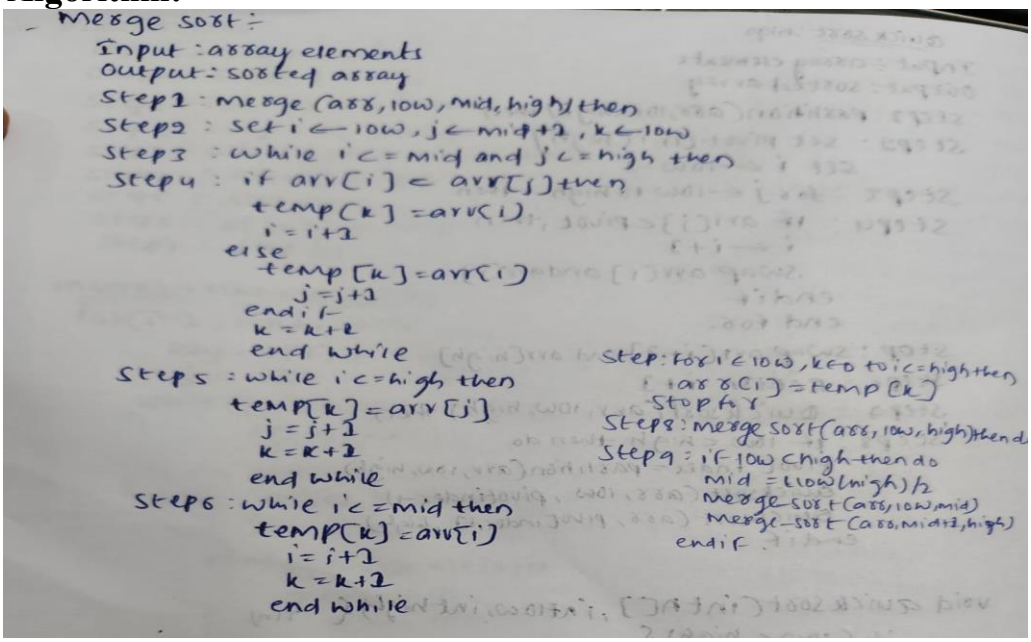
```
Enter the number of elements: 5
Enter 5 integers: 3
55
89
20
6
Sorted array: 3 6 20 55 89
```

Time complexity:



Merge Sort:

Algorithm:



Code:

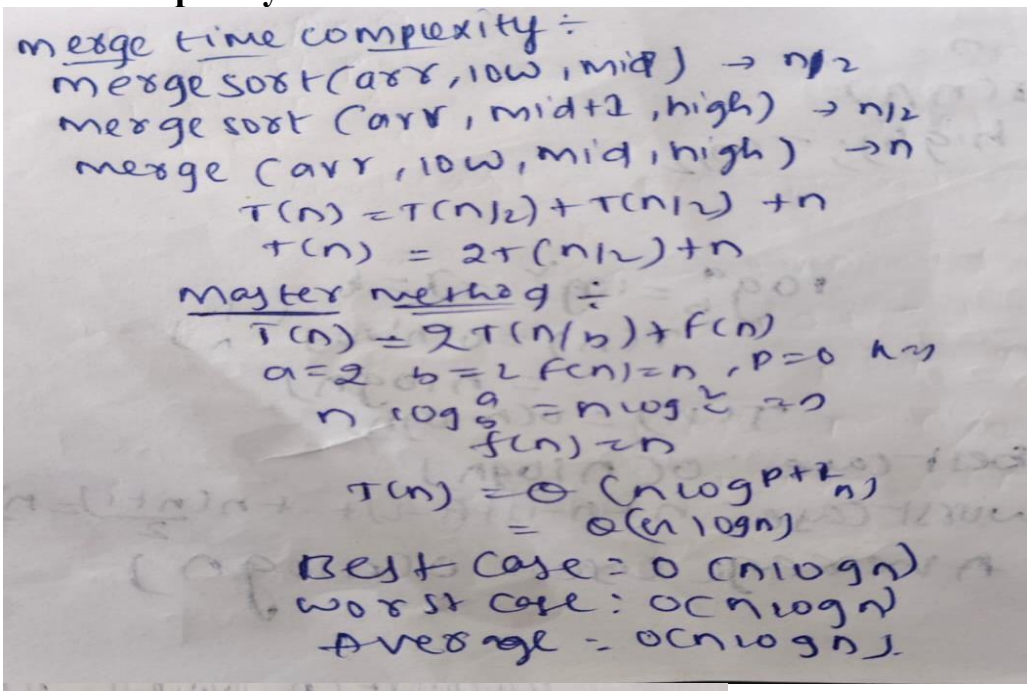
```
#include <iostream>
#include <vector>
void merge(std::vector<int>& arr, int left, int mid, int right) {
    int n1 = mid - left + 1;
    int n2 = right - left;
    std::vector<int> L(n1);
    std::vector<int> R(n2);
    for (int i = 0; i < n1; ++i) {
        L[i] = arr[left + i];}
    for (int j = 0; j < n2; ++j) {
        R[j] = arr[mid + 1 + j];
    int i = 0;
    int j = 0;
    int k = left;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) {
            arr[k] = L[i];
            ++i;
        } else {
            arr[k] = R[j];
            ++j;}++k;
```

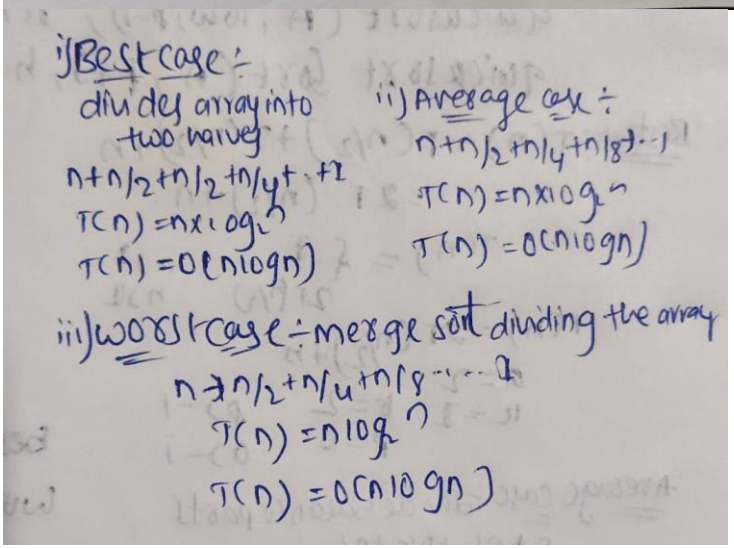
```
    while (i < n1) {
        arr[k] = L[i];
        ++i;
        ++k;
    }
    while (j < n2) {
        arr[k] = R[j];
        ++j;
        ++k;
    }
}

void mergeSort(std::vector<int>& arr, int left, int right) {
    if (left < right) {
        int mid = left + (right - left) / 2;

        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);

        merge(arr, left, mid, right);
    }
}

int main() {
    int n;
    std::cout << "Enter the number of elements: ";
    std::cin >> n;

    if (n <= 0) {
        return 0;
    }

    std::vector<int> arr(n);
    std::cout << "Enter " << n << " integers: ";
    for (int i = 0; i < n; ++i) {
        std::cin >> arr[i];
    }
    mergeSort(arr, 0, n - 1);
    std::cout << "Sorted array: ";
    for (int i = 0; i < n; ++i) {
        std::cout << arr[i] << " ";}
    std::cout << std::endl;
    return 0;
    }
```

Output:

```
Output
Enter the number of elements: 5
Enter 5 integers: 66
4
7
45
5
Sorted array: 4 5 7 45 66
```

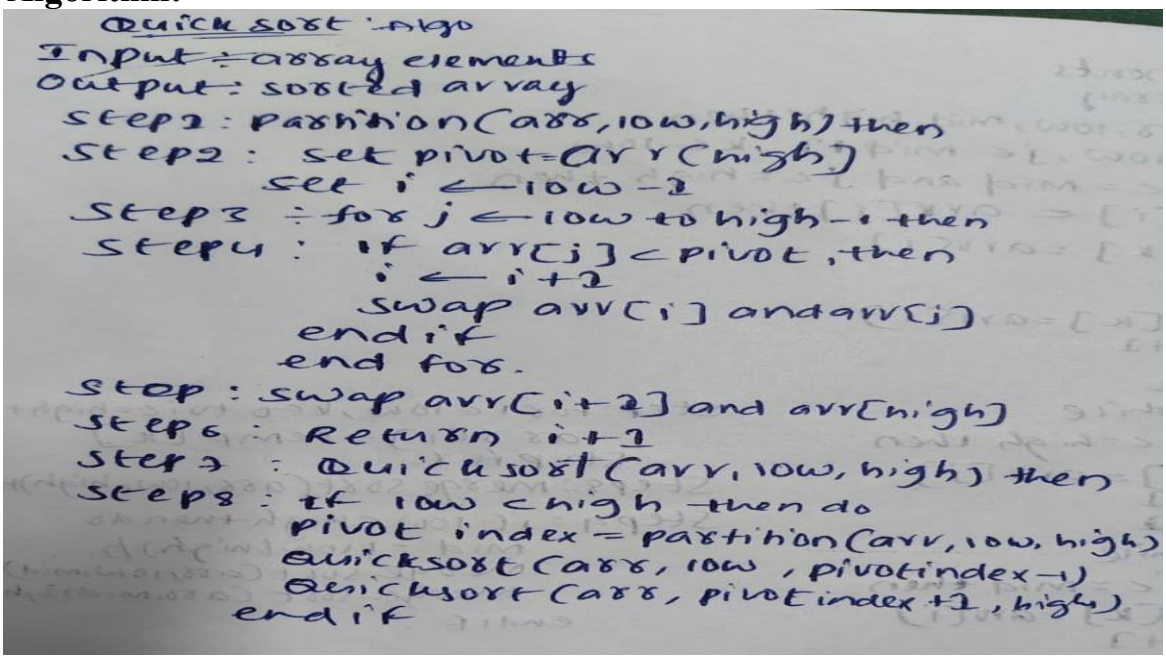
Time Complexity





Quicksort:

Algorithm:

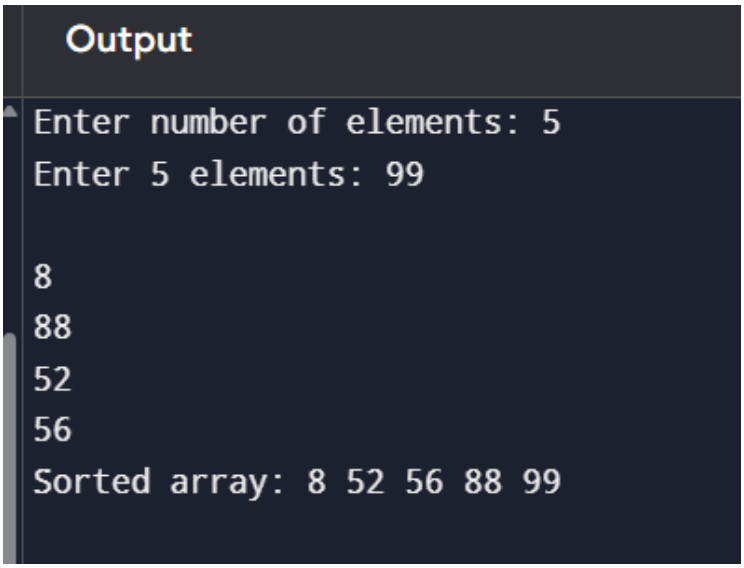


Code:

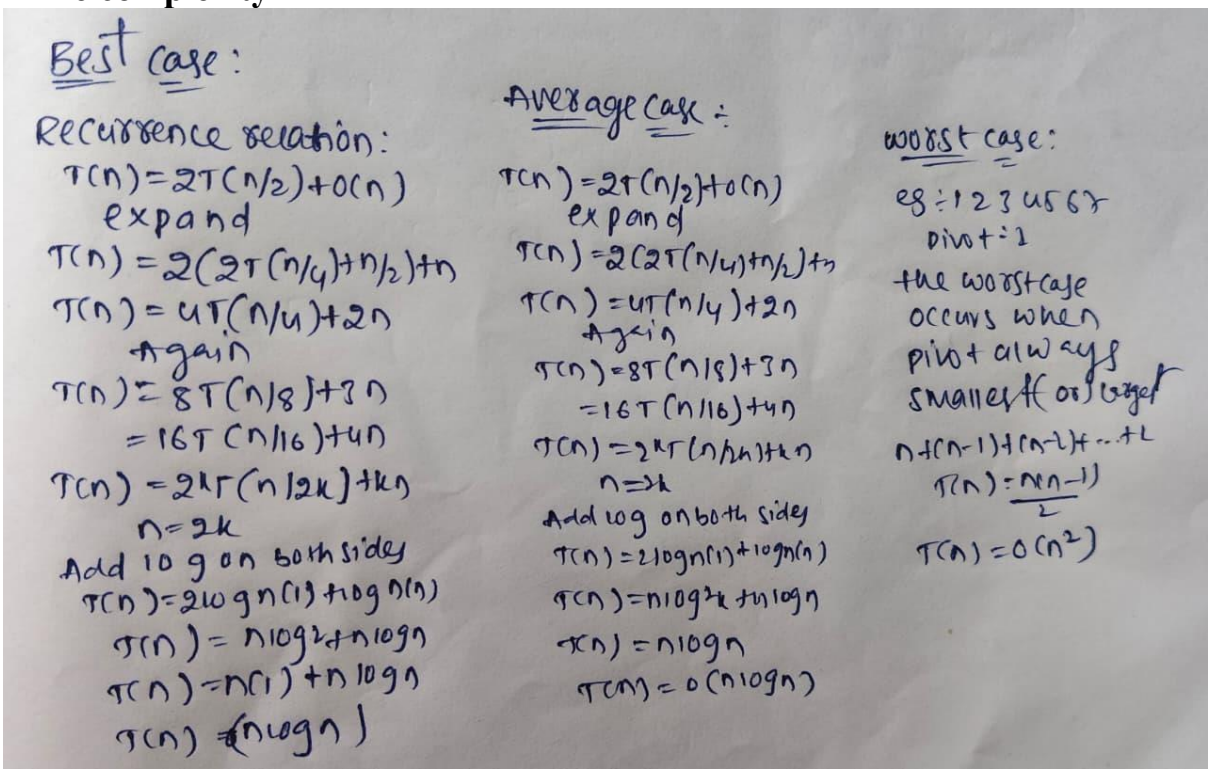
```
#include <iostream>
#include <vector>
void swap(int& a, int& b) {
    int temp = a;
    a = b;
    b = temp;
}
int partition(std::vector<int>& arr, int low, int high) {
    int pivot = arr[high];
    int i = low - 1;
    for (int j = low; j < high; j++) {
        if (arr[j] < pivot) {
            i++;
            swap(arr[i], arr[j]);}}
    swap(arr[i + 1], arr[high]);
    return i + 1;}
void quickSort(std::vector<int>& arr, int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);}}
int main() {
```

```
    std::cout << "Enter number of elements: ";
    std::cin >> n;

    std::vector<int> arr(n);
    std::cout << "Enter " << n << " elements: ";
    for (int i = 0; i < n; i++) {
        std::cin >> arr[i];
    }
    quickSort(arr, 0, n - 1);
    std::cout << "Sorted array: ";
    for (int i = 0; i < n; i++) {
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
    return 0;
}
```

Output:



Time complexity



Conclusion:

The implementation and time analysis of Bubble Sort, Selection Sort, Insertion Sort, MergeSort, and Quick Sort helps us understand how different sorting algorithms perform for different input sizes. By analyzing their best, average, and worst cases, we can choose the most suitable algorithm for a particular problem. Merge Sort and Quick Sort are generally faster for large datasets, while Bubble,Selection, and Insertion Sort are simpler and useful for small datasets.